A11 Box - Wire Cutting List

United Control Box, wiring rev 0.7 · 2026-10-31
From the 3D duct routing of the simulator (measured parts, 730 × 930 plate)

Cut = routed + 10 % + 15 cm, rounded up to 5 cm. 102 pieces, **92.9 m** in total. Door wires include 0.30 m for the hinge service loop. Multi-core cables are cut once, to the longest core. Routes follow the duct centre lines, so the real length can differ by ±15-20 %: cut the long runs a little long and trim on fitting. (2×) = twin ferrule with another wire on that screw.

Summary by wire type

Wire / cable	Pieces	Routed (m)	To cut (m)
1 mm² yellow, band dark blue	37	25.57	34.50
1 mm² yellow, band black	11	7.68	10.35
1 mm² yellow, band blue/white	14	7.16	10.25
2.5 mm² G/Y	6	6.68	8.40
4 mm² G/Y	5	2.91	4.10
4 mm² black	7	2.44	3.90
2.5 mm² blue	2	2.22	2.80
2.5 mm² black	2	1.97	2.50
1 mm² yellow, band blue	4	1.74	2.60
2.5 mm² brown, band dark blue	2	1.68	2.20
6 mm² braided strap	1	1.10	1.40
4 mm² blue	2	0.78	1.25
1 mm² yellow, band red	2	0.77	1.20
Motor cable X - LiYCY 4 × 1.0 shielded (A2 → XM-X)	1	0.89	1.15
Motor cable Y - LiYCY 4 × 1.0 shielded (A3 → XM-Y)	1	0.77	1.00
Motor cable Z - LiYCY 4 × 1.0 shielded (A4 → XM-Z)	1	0.72	0.95
PUL / DIR set X - 3 × 1 mm² yellow, twisted together (A11 → A2)	1	0.70	0.95
PUL / DIR set Y - 3 × 1 mm² yellow, twisted together (A11 → A3)	1	0.79	1.05
PUL / DIR set Z - 3 × 1 mm² yellow, twisted together (A11 → A4)	1	0.97	1.25
Spindle cable W1 - 4G1.5 shielded (VFD → HDC-HA)	1	0.79	1.05
Total	**102**		**92.85**

1 mm² yellow, band dark blue - 37 pieces, 34.50 m

	No.	From	To	Routed	Cut (m)	Door
☐	166	F2 out (2×)	A11 DC24V	1.11	**1.40**	
☐	168	F2 out (2×)	A11 24V (2×)	1.11	**1.40**	
☐	170	Contactor 43 (2×)	Fan +	1.08	**1.35**	
☐	201	F1 out (2×)	E-stop 1 (2×)	1.56	**1.90**	door
☐	202	E-stop 2	OFF 1	0.16	**0.35**	door
☐	203	OFF 2 (2×)	ON 1	0.07	**0.25**	door
☐	204	ON 2	K1 A1 (2×)	1.39	**1.70**	door
☐	205	OFF 2 (2×)	K1 11	1.52	**1.85**	door
☐	206	K1 14	K1 A1 (2×)	0.57	**0.80**	
☐	210	E-stop 4	A11 X7	1.68	**2.00**	door
☐	220	F1 out (2×)	Contactor 43 (2×)	0.81	**1.05**	
☐	221	Contactor 44	Yellow lamp +	1.40	**1.70**	door
☐	223	E-stop 1 (2×)	Green lamp + (2×)	0.31	**0.50**	door
☐	224	Green lamp -	A11 Y6	1.64	**2.00**	door
☐	225	Green lamp + (2×)	Red lamp +	0.13	**0.30**	door
☐	226	Red lamp -	A11 Y5	1.68	**2.05**	door
☐	304	DMA860H-X PUL+ (2×)	DMA860H-X DIR+	0.02	**0.20**	
☐	308	DMA860H-Y PUL+ (2×)	DMA860H-Y DIR+	0.02	**0.20**	
☐	312	DMA860H-Z PUL+ (2×)	DMA860H-Z DIR+	0.02	**0.20**	
☐	315	DMA860H-X ALM+ (2×)	A11 X5	0.54	**0.75**	
☐	317	DMA860H-Y ALM+ (2×)	DMA860H-X ALM+ (2×)	0.35	**0.55**	
☐	319	DMA860H-Z ALM+	DMA860H-Y ALM+ (2×)	0.35	**0.55**	
☐	325	A11 Y1	VFD X1	0.50	**0.70**	
☐	327	A11 Y2	VFD X2	0.50	**0.70**	
☐	328	A11 Y3	VFD X3	0.51	**0.75**	
☐	329	A11 Y4	VFD X4	0.52	**0.75**	
☐	330	X-LIM 1	A11 X1	0.47	**0.70**	
☐	331	Y-LIM 1	A11 X2	0.54	**0.75**	

	No.	From	To	Routed	Cut (m)	Door
☐	332	Z-LIM 2	A11 X3	0.57	**0.80**	
☐	334	X-LIM 2 (2×)	Y-LIM 2 (2×)	0.21	**0.40**	
☐	335	Y-LIM 2 (2×)	Z-LIM 1	0.23	**0.40**	
☐	336	X-LIM 2 (2×)	A11 X6	0.50	**0.75**	
☐	340	A11 24V (2×)	X-LIM 4 (2×)	0.52	**0.75**	
☐	341	X-LIM 4 (2×)	Y-LIM 4 (2×)	0.19	**0.40**	
☐	342	Y-LIM 4 (2×)	Z-LIM 4	0.19	**0.40**	
☐	345	Tool setter 1	A11 X4	0.62	**0.85**	
☐	347	Tool setter 3	E-stop 3	1.97	**2.35**	door

1 mm² yellow, band black - 11 pieces, 10.35 m

	No.	From	To	Routed	Cut (m)	Door
☐	112	CB-24V 2 (2×)	PSU 24V L	0.27	**0.45**	
☐	118	CB-48V 2	PSU 48V L	0.89	**1.15**	
☐	121	CB-PUMP 2	Schuko L	0.20	**0.40**	
☐	128	VFD P+	Braking resistor T1	1.00	**1.25**	
☐	129	VFD PB	Braking resistor T2	0.73	**1.00**	
☐	150	PSU 48V +V1	DMA860H-X AC1	0.75	**1.00**	
☐	151	PSU 48V -V1	DMA860H-X AC2	0.69	**0.95**	
☐	152	PSU 48V +V2	DMA860H-Y AC1	0.85	**1.10**	
☐	153	PSU 48V -V2	DMA860H-Y AC2	0.79	**1.05**	
☐	154	PSU 48V +V3	DMA860H-Z AC1	0.74	**1.00**	
☐	155	PSU 48V -V3	DMA860H-Z AC2	0.77	**1.00**	

1 mm² yellow, band blue/white - 14 pieces, 10.25 m

	No.	From	To	Routed	Cut (m)	Door
☐	167	PSU 24V -V1 (2×)	A11 DCGND (2×)	0.71	**0.95**	
☐	169	PSU 24V -V2 (2×)	A11 GND (2×)	0.73	**1.00**	
☐	171	A11 DCGND (2×)	Fan -	0.62	**0.85**	
☐	207	PSU 24V -V1 (2×)	K1 A2	0.72	**0.95**	
☐	222	K1 A2 (2nd screw)	Yellow lamp -	1.43	**1.75**	door
☐	316	A11 OGND (2×)	DMA860H-X ALM- (2×)	0.51	**0.75**	
☐	318	DMA860H-X ALM- (2×)	DMA860H-Y ALM- (2×)	0.36	**0.55**	
☐	320	DMA860H-Y ALM- (2×)	DMA860H-Z ALM-	0.32	**0.50**	
☐	326	A11 OGND (2×)	VFD COM	0.58	**0.80**	
☐	337	A11 GND (2×)	X-LIM 3 (2×)	0.53	**0.75**	
☐	338	X-LIM 3 (2×)	Y-LIM 3 (2×)	0.23	**0.40**	
☐	339	Y-LIM 3 (2×)	Z-LIM 3 (2×)	0.23	**0.40**	
☐	346	Z-LIM 3 (2×)	Tool setter 2 (2×)	0.18	**0.40**	
☐	348	Tool setter 2 (2×)	Tool setter 4	0.02	**0.20**	

2.5 mm² G/Y - 6 pieces, 8.40 m

	No.	From	To	Routed	Cut (m)	Door
☐	144	VFD PEc	Earth bar 1	0.90	**1.15**	
☐	145	PSU 48V PE	Earth bar 2	1.04	**1.30**	
☐	146	PSU 24V PE	Earth bar 2	1.29	**1.60**	
☐	147	DIN-rail earth stud	Earth bar 2	1.34	**1.65**	
☐	148	PSU 24V -V2 (2×)	Earth bar 2	0.92	**1.20**	
☐	149	Schuko PE	Earth bar 2	1.19	**1.50**	

4 mm² G/Y - 5 pieces, 4.10 m

	No.	From	To	Routed	Cut (m)	Door
☐	138	Earth bar 1	Earth bar 2	0.38	**0.60**	
☐	139	Enclosure body (earth stud)	Earth bar 1	0.56	**0.80**	
☐	140	Mains (gland) PE	Earth bar 1	0.43	**0.65**	
☐	141	Backplate	Earth bar 1	0.27	**0.45**	
☐	143	EMI filter GND	Earth bar 1	1.28	**1.60**	

4 mm² black - 7 pieces, 3.90 m

	No.	From	To	Routed	Cut (m)	Door
☐	103	Main CB 2	EMI filter LIN	0.14	**0.30**	
☐	105	EMI filter LOUT (2×)	Contactor 1L1	0.23	**0.45**	
☐	109	Contactor 2T1	L busbar XL	0.52	**0.75**	
☐	111	EMI filter LOUT (2×)	CB-24V 1	0.33	**0.55**	
☐	114	L busbar XL	CB-VFD 1	0.37	**0.60**	
☐	117	L busbar XL	CB-48V 1	0.40	**0.60**	
☐	120	L busbar XL	CB-PUMP 1	0.44	**0.65**	

2.5 mm² blue - 2 pieces, 2.80 m

	No.	From	To	Routed	Cut (m)	Door
☐	102	Mains (gland) N	Main CB 3	1.23	**1.55**	
☐	116	N busbar XN	VFD N	0.99	**1.25**	

2.5 mm² black - 2 pieces, 2.50 m

	No.	From	To	Routed	Cut (m)	Door
☐	101	Mains (gland) L	Main CB 1	1.20	**1.50**	
☐	115	CB-VFD 2	VFD L	0.77	**1.00**	

1 mm² yellow, band blue - 4 pieces, 2.60 m

	No.	From	To	Routed	Cut (m)	Door
☐	113	N busbar XN	PSU 24V N	0.35	**0.55**	
☐	119	N busbar XN	PSU 48V N	0.71	**0.95**	
☐	122	N busbar XN	Schuko N	0.25	**0.45**	
☐	132	N busbar XN	Contactor A1	0.43	**0.65**	

2.5 mm² brown, band dark blue - 2 pieces, 2.20 m

	No.	From	To	Routed	Cut (m)	Door
☐	162	PSU 24V +V1	F1 in	0.87	**1.15**	
☐	164	PSU 24V +V2	F2 in	0.80	**1.05**	

6 mm² braided strap - 1 piece, 1.40 m

	No.	From	To	Routed	Cut (m)	Door
☐	142	Door strap	Earth bar 1	1.10	**1.40**	door

4 mm² blue - 2 pieces, 1.25 m

	No.	From	To	Routed	Cut (m)	Door
☐	104	Main CB 4	EMI filter NIN	0.14	**0.35**	
☐	106	EMI filter NOUT	N busbar XN	0.64	**0.90**	

1 mm² yellow, band red - 2 pieces, 1.20 m

	No.	From	To	Routed	Cut (m)	Door
☐	130	CB-24V 2 (2×)	K1 41	0.62	**0.85**	
☐	131	K1 44	Contactor A2	0.15	**0.35**	

Multi-core cables (cut once per cable)

	Cable	Cores (no.: from → to)	Longest	Cut (m)
☐	Motor cable X - LiYCY 4 × 1.0 shielded (A2 → XM-X)	DMA860H-X A+ → XM-X 1 (0.87) DMA860H-X A- → XM-X 2 (0.89) DMA860H-X B+ → XM-X 3 (0.89) DMA860H-X B- → XM-X 4 (0.89)	0.89	**1.15**
☐	Motor cable Y - LiYCY 4 × 1.0 shielded (A3 → XM-Y)	DMA860H-Y A+ → XM-Y 1 (0.77) DMA860H-Y A- → XM-Y 2 (0.76) DMA860H-Y B+ → XM-Y 3 (0.76) DMA860H-Y B- → XM-Y 4 (0.76)	0.77	**1.00**
☐	Motor cable Z - LiYCY 4 × 1.0 shielded (A4 → XM-Z)	DMA860H-Z A+ → XM-Z 1 (0.72) DMA860H-Z A- → XM-Z 2 (0.71) DMA860H-Z B+ → XM-Z 3 (0.70) DMA860H-Z B- → XM-Z 4 (0.71)	0.72	**0.95**
☐	PUL / DIR set X - 3 × 1 mm² yellow, twisted together (A11 → A2)	301: A11 X_5V → DMA860H-X PUL+ (2×) (0.62) 302: A11 X_PUL → DMA860H-X PUL- (0.62) 303: A11 X_DIR → DMA860H-X DIR- (0.70)	0.70	**0.95**
☐	PUL / DIR set Y - 3 × 1 mm² yellow, twisted together (A11 → A3)	305: A11 Y_5V → DMA860H-Y PUL+ (2×) (0.79) 306: A11 Y_PUL → DMA860H-Y PUL- (0.79) 307: A11 Y_DIR → DMA860H-Y DIR- (0.79)	0.79	**1.05**
☐	PUL / DIR set Z - 3 × 1 mm² yellow, twisted together (A11 → A4)	309: A11 Z_5V → DMA860H-Z PUL+ (2×) (0.88) 310: A11 Z_PUL → DMA860H-Z PUL- (0.97) 311: A11 Z_DIR → DMA860H-Z DIR- (0.97)	0.97	**1.25**

	Cable	Cores (no.: from → to)	Longest	Cut (m)
☐	Spindle cable W1 - 4G1.5 shielded (VFD → HDC-HA)	123: VFD U → HDC-HA 1 (0.75) 124: VFD V → HDC-HA 2 (0.76) 125: VFD W → HDC-HA 3 (0.77) 126: VFD PE → HDC-HA PE (0.79) 127: HDC-HA SH → VFD PEb (0.78)	0.79	**1.05**

Not in this list: 12 conductors outside the box (motor and sensor cables from the connectors to the machine) - their length depends on the machine.
Source: Verification/A11_Wire_Lengths_rev07.csv; regenerate with make_wire_lengths.py after re-exporting the lengths (player -a11lengths).